PAPER_227: Tapestry of Blazing Starbirth (NGC 2014/2020, LMC) — MUGE with Gas-Accreting Mass M(t) and Stellar Wind Ram Pressure

Author: Daniel T. Murphy **Framework:** UQFF v4.8 (Star-Magic) **Session:** 58 (grok_share_8d951e12.txt extraction — Doc 4) **Date:** March 2026 **Classification:** Novel MUGE — Gas-Ratio Amplitude M(t) + Stellar Wind Ram Pressure **Status:** Proof-Quality Whitepaper

---

Abstract

The Tapestry of Blazing Starbirth (NGC 2014 and NGC 2020) in the Large Magellanic Cloud is modelled using a 9-term MUGE incorporating two novel mathematical methods: (1) time-varying stellar mass $M(t) = M_{init}(1 + M_{dot\_factor} \cdot e^{-t/\tau_{SF}})$ where the amplitude $M_{dot\_factor} = M_{gas}/M_{init} \approx 41.7$ encodes the gas-to-stellar mass ratio, and (2) stellar wind ram-pressure acceleration $a_{wind} = \rho_{wind} v_{wind}^2 / \rho_{fluid}$. At canonical parameters $a_{wind} \approx 4 \times 10^3$ m/s^2 dominates most other MUGE terms during active O/B-star formation.

---

1. Physical System

NGC 2014 and NGC 2020 are two giant H II regions in the Large Magellanic Cloud, separated by ~300 ly, forming the "Tapestry of Blazing Starbirth" (HST WFC3 image, 2020). Key parameters:

Parameter	Value
Distance	~160,000 ly (LMC)
M_{init}	$240 M_\odot$ (stellar; young massive stars)
M_{gas}	$10{,}000 M_\odot$ (surrounding gas)
M_{dot_factor}	$M_{gas}/M_{init} = 41.7$
r	10 ly
B	1 μT
z	LMC ≈ -0.0005 (Milky Way satellite)
τ_{SF}	5 Myr

---

2. Novel Equations

2.1 Gas-Ratio-Amplitude Mass Growth

$$M(t) = M_{init} \left(1 + \frac{M_{gas}}{M_{init}} e^{-t/\tau_{SF}} \right)$$

The amplitude M_{gas}/M_{init} is the gas-to-stellar mass ratio, here ≈ 41.7. This captures the rapid initial mass increase as gas accretes onto young stellar objects. By $t = 5\tau_{SF}$, M returns to M_{init}.

2.2 Stellar Wind Ram Pressure Acceleration

$$a_{wind} = \frac{\rho_{wind} \cdot v_{wind}^2}{\rho_{fluid}}$$

Parameter	Value
ρ_{wind}	10^{-21} kg/m^3 (O/B-star wind)
v_{wind}	2000 km/s = 2×10^6 m/s
ρ_{fluid}	10^{-21} kg/m^3 (ambient medium)

At these parameters: $a_{wind} = (10^{-21} \times 4 \times 10^{12})/10^{-21} = 4 \times 10^{12}$ m/s^2.

---

3. Context Within MUGE Stellar Wind Family

System	ρ_{wind} (kg/m^3)	v_{wind} (km/s)	ρ_{fluid} (kg/m^3)
Tapestry LMC	10^{-21}	2000	10^{-21}
Westerlund 2	10^{-20}	2000	10^{-21}
NGC 1792 (SN)	10^{-21}	2000	10^{-21}

Westerlund 2 has ρ_{wind} 10× higher, distinguishing it as the denser OB-supergiant environment.

---

4. Simulation Parameters

Parameter	Value
$t_{canonical}$	1 Myr
dt	0.01 Myr
τ_{SF}	5 Myr
E_0 erosion	No erosion term (use PAPER_229 for Pillars)

---

5. Calculator Class

```python
class StarbirthTapestryLMCUQFFCalculator(_CP3Calculator):
    """PAPER_227: NGC 2014/2020 LMC Tapestry — 9-term MUGE, gas-ratio M(t), stellar win
    # Session 58 — grok_share_8d951e12.txt Doc 4
```

Located in `CondensedPhysics3.py` (Session 58).

---

6. Conclusion

The Tapestry of Blazing Starbirth introduces two novel MUGE methods: gas-ratio-amplitude mass evolution and stellar wind ram-pressure acceleration. The a_{wind} term, at $\rho_{wind} = \rho_{fluid}$, evaluates to v_{wind}^2 — numerically dominant during the O/B-star-active phase. This establishes a parametric wind family within the LMC/Milky Way comparison (see PAPER_228 for the 10× denser Westerlund 2 case).

Source: grok_share_8d951e12.txt — Doc 4 (Tapestry Starbirth LMC MUGE)

UQFF computed: UQFF magnetic Jeans correction factor [SSq]B/(8p?c_s) = 5.7e-1 × 1.3e-9 = 7.4e-10; Jeans mass deviation from standard = 7.4e-10 M_J.